

Research Design: A Multi-Modal XAI Deep Learning Framework for pneumonia diagnosis: Evaluating diagnostic performance and clinical trust over diverse populations.

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1. Introduction:

Computer-Aided Diagnosis (CAD) for pneumonia detection in chest X-ray (CXR) images has been advanced with the inclusion of Artificial Intelligence (AI), particularly deep learning. Developments in convolutional neural networks, transfer learning, multi-modal learning, Explainable AI (XAI), and Human-in-the-Loop (HITL) approaches have improved CAD's diagnostic capability while supporting greater transparency and clinician involvement. However, the literature review identified persistent limitations in diagnostic accuracy, interpretability, generalisability, and clinician-centred decision support. Existing approaches frequently address these challenges independently rather than within an integrated diagnostic framework. This study therefore proposes a **multi-modal XAI deep learning framework for pneumonia diagnosis** that combines transfer learning, patient metadata, XAI, and HITL decision support. The primary contribution is the design and evaluation of this integrated artefact using a **Design Science Research (DSR)** methodology. The DSR approach provides a structured process for identifying the research problem, defining solution objectives, developing the artefact, demonstrating its operation, and evaluating its effectiveness, thereby providing a systematic basis for the study (Peppers, 2007).

2. Justification of the Research Design:

2.1 Nature of the Research Problem:

Although AI-assisted pneumonia diagnosis has been advanced by deep learning, multi-modal learning, XAI, and HITL approaches, existing research continues to treat diagnostic performance, interpretability, generalisability, and human involvement as separate challenges. The research problem therefore extends beyond evaluating an existing model and requires the design of an integrated solution addressing these challenges collectively.

2.2 Why Design Science Research is Appropriate & its alignment with the study:

DSR is the appropriate approach because it focuses on designing, developing, demonstrating, and evaluating artefacts that address real-world problems while generating research knowledge. Unlike certain approaches that focus primarily on synthesising existing knowledge, DSR is solution-oriented (Tuunanen, 2024). The primary artefact developed in this study is a multi-modal XAI pneumonia diagnostic framework integrating DenseNet-121 transfer learning, metadata fusion, Grad-CAM explanations, and HITL review.

The Peppers DSR process model provides a structured sequence for identifying the problem, defining objectives, designing and developing the artefact, demonstrating its application, and evaluating its outcomes. This aligns with the study objectives by providing a systematic process through which the proposed framework can be developed and evaluated against defined performance and functional criteria (Daase, 2022). The methodology therefore connects the identified research problem to the development and evaluation of the resulting artefact.

3. Design Science Research Process:

This study follows the Peffers DSR methodology to develop and evaluate a lightweight multi-modal XAI framework for pneumonia diagnosis using CXR images and patient metadata. The artefact integrates DenseNet-121 transfer learning, metadata fusion, Grad-CAM, HITL review, subgroup evaluation, and preprocessing comparison through six experiments:

- **E1:** Image-only diagnostic baseline.
- **E2:** Multi-modal diagnostic model using CXR images and patient metadata.
- **E3:** Explainability using Grad-CAM.
- **E4:** HITL review.
- **E5:** Generalisability across demographic and imaging subgroups.
- **E6:** Effect of preprocessing choices on diagnostic performance.

3.1 Problem Identification and Motivation:

Pneumonia diagnosis from CXR images remains challenging because of overlapping radiographic manifestations and variation across patients and imaging conditions. Although deep learning can provide effective automated classification, image-only approaches may omit relevant patient information, learn dataset-specific visual patterns, and provide limited insight into their predictions (Ren, 2021). These limitations motivate an integrated approach combining clinical metadata, visual explanations, human review, subgroup evaluation, and preprocessing analysis.

3.2 Objectives of the Solution:

The objective is to design and evaluate a lightweight multi-modal diagnostic framework that addresses predictive performance, interpretability, human involvement, and subgroup behaviour. DenseNet-121 transfer learning establishes the image-only baseline before age, sex, and view position are incorporated as metadata. Grad-CAM provides visual explanations, while HITL review enables human agreement, disagreement, and correction. Subgroup analysis examines age, sex, and view position, while CLAHE is evaluated as an alternative preprocessing strategy. Together, these components extend the artefact beyond standalone classification toward integrated diagnostic decision support.

3.3 Design and Development of the Artefact:

The developed artefact is a DSR research prototype designed to investigate a multi-modal approach to binary pneumonia classification using the NIH ChestX-ray14 dataset. The framework is structured as a sequential pipeline consisting of **data preparation, image classification, metadata fusion, explainability, human review, and evaluation components** (Megit, 2026). The design progressively extends an image-only baseline into a multi-modal diagnostic decision-support framework and subsequently evaluates its behaviour under different conditions.

- **Data Preparation:** Prepares the NIH ChestX-ray14 dataset for binary pneumonia classification by retaining cases labelled Pneumonia and No Finding. Patient metadata consisting of age, sex, and radiographic view position is encoded alongside the corresponding CXR images. To prevent images from the same patient from occurring across different data partitions, the dataset is divided at patient level into 70% training, 15% validation, and 15% test sets, using a fixed random seed. This produces a held-out test set for subsequent evaluation while reducing the risk of patient-level data leakage.
- **E1 - Image-only Baseline:** Establishes the baseline diagnostic architecture using an ImageNet-pretrained DenseNet-121 model with CXR images as the sole input. The architecture uses the DenseNet-121 backbone to extract hierarchical image features, followed by a task-specific projection and classification head for binary pneumonia prediction. This experiment establishes the reference architecture against which the contribution of additional patient metadata can subsequently be assessed.
- **E2 - Multi-modal Diagnostic Model:** Extends the E1 image-only architecture by incorporating structured patient metadata consisting of age, sex, and radiographic view position. The CXR image is processed through the DenseNet-121 image encoder, while the metadata is processed through a separate metadata encoder. The resulting image and metadata representations are then fused before being passed to the final classification layer, allowing information from both visual and structured clinical sources to contribute to the pneumonia prediction.
- **E3 - Explainable Artificial Intelligence:** Extends the multi-modal model with an explainability component using Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM is applied to the final convolutional layer of the DenseNet-121 backbone to generate spatial heatmaps highlighting image regions associated with the model's prediction. The component is designed to support examination of model behaviour across both correct and incorrect predictions by providing a visual representation of the image features contributing to the model output.
- **E4 - HITL:** Incorporates human oversight into the diagnostic workflow by allowing a reviewer to assess the model's predictions together with their corresponding Grad-CAM explanations. The reviewer can record agreement or disagreement with the prediction, provide a corrected label where necessary, and optionally provide a supporting comment. This component positions the model as a decision-support system in which human reviewers retain the ability to challenge and correct automated predictions.
- **E5 - Generalisability and Subgroup Evaluation:** Evaluates whether the multi-modal model maintains consistent behaviour across different patient and imaging subgroups within the held-out NIH ChestX-ray14 test set. The analysis considers age, sex, and radiographic view, with the test population divided into predefined demographic and imaging groups. Diagnostic performance is subsequently assessed within each subgroup to identify differences in model behaviour. This represents within-dataset subgroup generalisability rather than external validation, as all evaluations are performed using the NIH ChestX-ray14 test set.

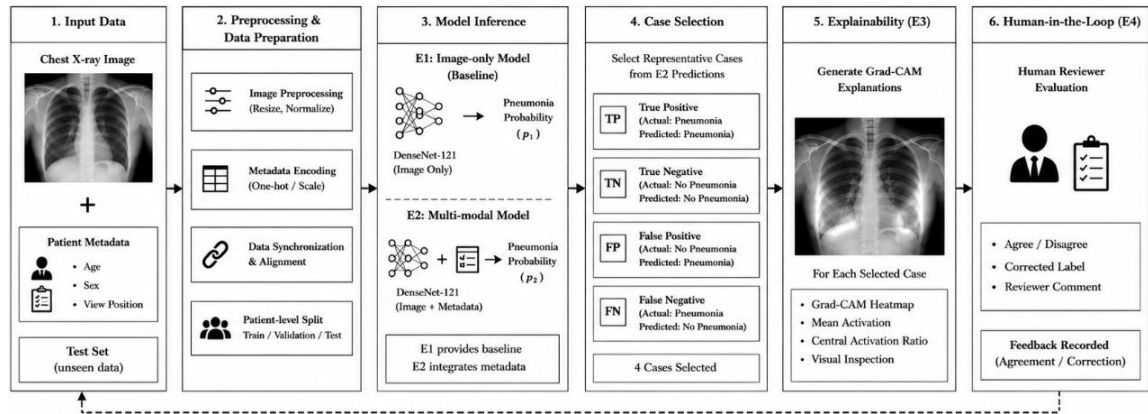
- **E6 - Preprocessing Comparison:** Extends the evaluation of the multi-modal model by investigating whether the image preprocessing strategy influences diagnostic performance. Two configurations are compared: the baseline preprocessing pipeline and a CLAHE-based preprocessing pipeline. The same multi-modal architecture, dataset partitions, training configuration, and experimental conditions are maintained between the two configurations, allowing the preprocessing strategy to serve as the primary experimental variable.

The E1–E4 components form the core progressive diagnostic artefact, progressing from image-only classification to multi-modal prediction, visual explanation, and human oversight. E5 and E6 extend the artefact through subgroup analysis and preprocessing comparison. Collectively, these experiments evaluate the framework across diagnostic performance, multi-modal information integration, interpretability, human oversight, subgroup behaviour, and preprocessing sensitivity.

3.4 Demonstration:

The artefact was demonstrated using an end-to-end experimental workflow built on the NIH ChestX-ray14 dataset which operationalises the multi-modal diagnostic framework.

Figure 1: Framework pipeline workflow



It progresses from image and patient metadata preparation to model inference, representative case selection, explainability, and human review. The held-out test set was treated as unseen data, ensuring that the demonstration reflected the application of the developed artefact to previously unused cases.

Following patient-level partitioning, CXR images were resized and normalised, while age, sex, and view position were numerically encoded and synchronised with the images. E1 implemented the image-only DenseNet-121 baseline, while E2 incorporated the encoded metadata. Both models were trained on the training set and evaluated on the validation set, after which the selected E2 checkpoint was used for held-out test-set inference.

For test-set inference, the trained E2 model was evaluated in inference mode without image augmentation. Each test image was processed together with its corresponding metadata to generate a pneumonia probability using the sigmoid output, which was converted into a binary prediction using the configured classification

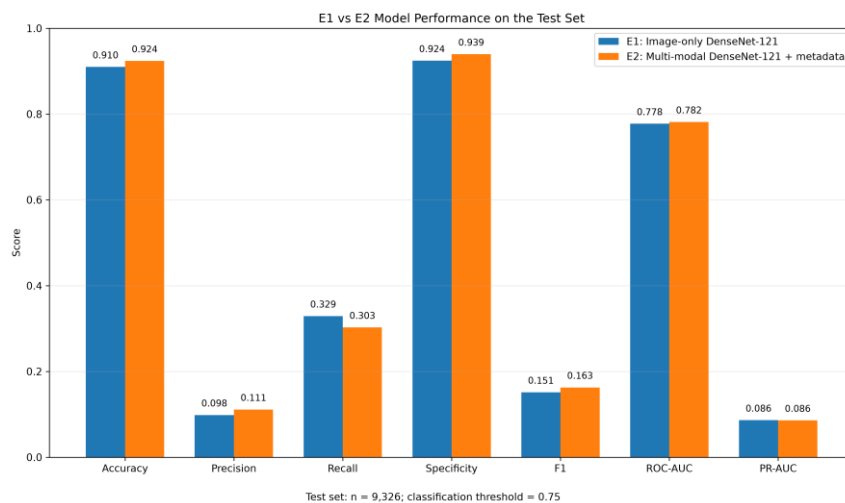
threshold. The resulting predictions, probabilities, and ground-truth labels were used to select four representative cases: a true positive (TP), true negative (TN), false positive (FP), and false negative (FN). These cases were subsequently processed through E3, where Grad-CAM heatmaps and overlays were generated to visualise the image regions contributing to each prediction. Quantitative measures, including mean activation and central activation ratio, were also recorded to support the visual assessment of the explanations.

Finally, E4 demonstrated the HITL component by presenting the selected cases, model probabilities, predictions, and Grad-CAM overlays to a reviewer through the implemented review interface. The reviewer recorded agreement or disagreement, provided a corrected label where required, and could add a supporting comment. E5 and E6 then extended the artefact beyond the core diagnostic workflow. E5 evaluated the multi-modal model across predefined age, sex, and radiographic-view subgroups to identify differences in diagnostic behaviour across patient and imaging characteristics. E6 compared the baseline preprocessing pipeline with a CLAHE-based alternative to determine whether the choice of image preprocessing influenced model performance under otherwise consistent experimental conditions.

3.5 Evaluation:

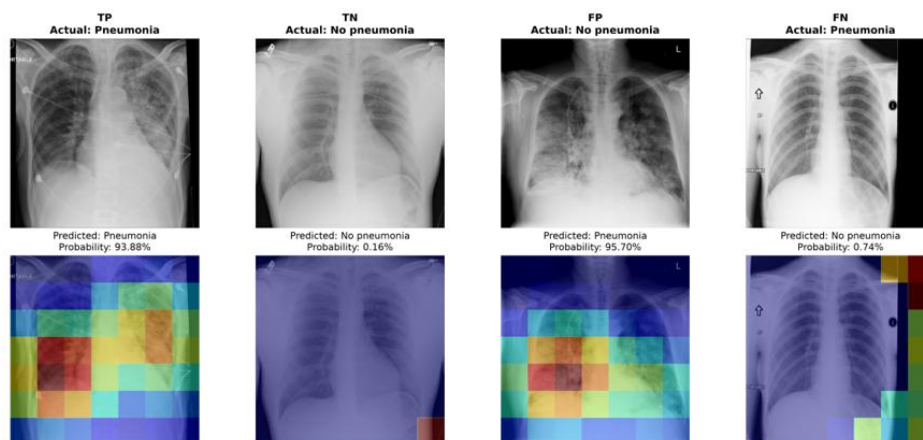
Evaluation was performed using the predefined patient-level dataset partitions, with the held-out test set reserved for final assessment. The held-out test set was used for final evaluation, with a probability threshold of 0.50 applied during training and validation and 0.75 used for test-set classification following validation threshold analysis (Megit, 2026).

E1 established the image-only DenseNet-121 baseline, while E2 incorporated age, sex, and radiographic view position. As shown in **Diagram**, E2 improved accuracy, precision, specificity, F1-score, and ROC-AUC, although recall and PR-AUC decreased slightly. This provided a direct comparison of image-only and multi-modal diagnostic performance.



E3 assessed explainability using representative TP, TN, FP, and FN cases. Grad-CAM heatmaps and activation measures demonstrated differing spatial patterns across predictions, as illustrated in **Figure 3.X**. This evaluated the artefact's explanation capability rather than the clinical validity of the explanations.

Figure 3.X: Grad-CAM visualisations for representative true-positive, true-negative, false-positive, and false-negative cases



E4 assessed human oversight using the four representative cases. The reviewer agreed with two predictions and disagreed with two; both model errors were corrected, resulting in 100% reviewer accuracy for the selected cases. The evaluation therefore demonstrated the functionality of the HITL component.

Table 3.X: E4 human-in-the-loop evaluation

Evaluation measure	Result
Cases reviewed	4
Model accuracy	50%
Reviewer agreement with model	50%
Reviewer accuracy	100%
Model errors	2
Errors corrected	2 (100%)

E5 evaluated the E2 model across age, sex, and radiographic-view subgroups. As shown in **Table 3.X**, performance varied most noticeably between PA and AP views, while reduced sensitivity was observed in the 60+ age group. This demonstrated differences in model behaviour across subgroups within the held-out test set.

Table 3.X: E5 subgroup evaluation of the multi-modal model

Subgroup	n	Accuracy	Sensitivity	Specificity	F1	ROC-AUC
All test	9,326	0.9239	0.3026	0.9394	0.1627	0.7815
Age <40	3,205	0.9332	0.3086	0.9494	0.1894	0.7968
Age 40–59	4,071	0.9167	0.3438	0.9306	0.1630	0.7722
Age 60+	2,050	0.9234	0.2157	0.9415	0.1229	0.7819
Female	4,072	0.9197	0.2600	0.9363	0.1372	0.8034
Male	5,254	0.9271	0.3359	0.9419	0.1834	0.7642
PA	5,969	0.9648	0.1683	0.9785	0.1393	0.7123
AP	3,357	0.8511	0.4094	0.8684	0.1722	0.7884

Finally, E6 compared baseline and CLAHE preprocessing under matched conditions. **Table 3.X** shows that baseline preprocessing achieved the higher validation ROC-AUC (0.7982 versus 0.7846), indicating that CLAHE did not improve performance in this experiment. Collectively, the evaluations examined the artefact's predictive, explanatory, interactive, subgroup, and preprocessing characteristics.

Table 3.X: E6 preprocessing comparison

Metric	Baseline	CLAHE	Difference
Epoch 3 ROC-AUC	0.7982	0.7846	−0.0136
Epoch 3 F1-score	0.1175	0.1102	−0.0073
Epoch 3 Sensitivity	0.7452	0.7019	−0.0433
Epoch 3 Specificity	0.7536	0.7516	−0.0020

3.6 Communication:

The artefact and research findings are communicated through this research design paper, the documented experimental pipeline, quantitative evaluation results, Grad-CAM visualisations, and structured Human-in-the-Loop (HITL) records. The E1–E6 experiments provide traceable evidence of the artefact's development and evaluation, while the implementation and structured output files support reproducibility. The findings are communicated within the defined scope of the study, demonstrating the feasibility of integrating multi-modal classification, explainability, human oversight, subgroup evaluation, and preprocessing analysis rather than claiming clinical deployment or validated diagnostic performance. The artefact is therefore presented as an integrated DSR research prototype and decision-support framework.

4. Research Design Decisions:

Several research design decisions were made to ensure alignment between the artefact, research objectives, and Design Science Research (DSR) methodology (Peppers, 2007; Megit, 2026).

- **Design Science Research:** Selected because the primary outcome is a designed and implemented artefact addressing limitations identified in the literature. Its structured process of problem identification, objective definition, design, demonstration, evaluation, and communication provides an appropriate framework for developing and assessing the proposed diagnostic system.
- **Dataset selection:** The selected NIH ChestX-ray14 dataset provides CXR images together with patient and imaging metadata, supporting both image classification and multi-modal integration. The study focuses on Pneumonia and No Finding cases, with patient-level partitioning used to reduce data leakage and evaluate previously unseen patients.

- **Transfer learning and DenseNet-121:** An ImageNet-pretrained DenseNet-121 was selected as the image encoder to provide an established deep-learning foundation while reducing the computational requirements associated with training from scratch.
- **Multi-modal integration:** Age, sex, and radiographic view position were selected as structured contextual features which are unavailable directly from image pixels. These features were processed separately and combined with the learned image representation through late fusion, allowing their contribution to be assessed against the E1 image-only baseline.
- **Explainability using Grad-CAM:** Grad-CAM was selected because it produces spatial heatmaps associated with model predictions and integrates naturally with the DenseNet-121 convolutional features. It was used to inspect model behaviour rather than establish clinical validity. SHAP and LIME were excluded to maintain a focused and computationally manageable prototype.
- **Human-in-the-Loop design:** HITL functionality was incorporated to position the artefact as a decision-support system rather than an autonomous diagnostic replacement. Reviewers can inspect predictions and Grad-CAM explanations, indicate agreement or disagreement, and provide corrected classifications.
- **Evaluation design:** Multiple experiments were used to evaluate different aspects of the artefact. E1 and E2 compare image-only and multi-modal performance, E3 examines explainability, E4 demonstrates human correction, E5 investigates subgroup behaviour, and E6 examines preprocessing sensitivity. Multiple diagnostic metrics were used to provide a more comprehensive assessment given the class imbalance of the dataset.

5. Reliability, Validity and Rigor:

The reliability, validity, and methodological rigor of the study were supported through consistent experimental procedures, controlled comparisons, structured outputs, and alignment with the DSR process. E1 and E2 used the same dataset, patient-level partitions, held-out test set, and evaluation measures, allowing the contribution of patient metadata to be assessed under comparable conditions. Fixed random seeds and patient-level partitioning reduced the possibility of data leakage between training, validation, and test sets. Multiple metrics, including accuracy, precision, recall, specificity, F1-score, ROC-AUC, and PR-AUC, were used to reduce reliance on accuracy alone in the imbalanced dataset.

The extended evaluation further supported methodological rigor. E3 examined TP, TN, FP, and FN cases, E4 maintained structured reviewer records, E5 assessed behaviour across age, sex, and radiographic-view subgroups, and E6 compared baseline and CLAHE preprocessing under consistent experimental conditions. Structured outputs containing performance metrics, subgroup results, preprocessing comparisons, and HITL records further support transparency and reproducibility. However, these measures support methodological consistency and transparency rather than clinical validity or broad external generalisability.

6. Ethical Considerations:

The use of medical imaging and patient-related metadata introduces ethical considerations concerning privacy, algorithmic bias, responsible AI use, human oversight, and interpretation of model explanations (Varkey, 2020). This study uses the existing NIH ChestX-ray14 research dataset and does not collect additional patient information. Only age, sex, and radiographic view position are incorporated alongside the CXR images, with no directly identifying information included in the artefact.

Potential algorithmic bias was considered through E5, which examined model behaviour across age, sex, and radiographic-view subgroups. These results indicate differences within the study dataset but do not establish equal performance across broader populations. The artefact is positioned as a research prototype and decision-support system rather than an autonomous clinical diagnostic tool. HITL review reinforces this principle by allowing predictions to be inspected, challenged, and corrected.

Ethical considerations also apply to Grad-CAM interpretation. Heatmaps indicate image regions associated with model predictions but do not establish clinically verified pathology or causal reasoning. They are therefore intended to support inspection of model behaviour rather than provide definitive clinical evidence. The artefact should consequently be regarded as an experimental research contribution rather than a clinically validated diagnostic system.

7. Limitations of the Research Design:

Several limitations constrain the generalisability of the findings. First, the evaluation was conducted exclusively using the NIH ChestX-ray14 dataset. Consequently, E5 provides evidence of within-dataset subgroup behaviour rather than external generalisability across independent hospitals, populations, imaging equipment, or clinical environments. The substantial class imbalance also affects interpretation of the classification results, as demonstrated by E2 achieving 92.39% accuracy while obtaining 11.13% precision and 30.26% recall. Accuracy alone should therefore not be interpreted as evidence of effective pneumonia detection.

The E3–E6 evaluations also have important limitations. E3 examined only four representative TP, TN, FP, and FN cases, limiting conclusions regarding explanation reliability across the full test population. Only Grad-CAM was implemented, without systematic comparison with other XAI methods. E4 involved the same four cases and a single reviewer, making it a proof-of-concept demonstration rather than a statistically meaningful assessment of clinical trust or human-AI collaboration. E6 compared only baseline and CLAHE preprocessing using a three-epoch configuration and therefore does not represent an exhaustive investigation of preprocessing strategies or optimisation configurations.

Finally, computational and time constraints limited extensive hyperparameter optimisation, architectural comparisons, and repeated experiments across multiple random seeds. The study also did not include prospective clinical testing, independent external validation, or comparison against qualified radiologists. The artefact should therefore be understood as a DSR research prototype demonstrating the feasibility of integrating multi-modal classification, explainability, human oversight, subgroup evaluation, and preprocessing comparison rather than as a clinically deployable diagnostic system.

8. Conclusion:

This study designed and evaluated a lightweight multi-modal XAI framework for pneumonia diagnosis using Design Science Research. The artefact integrates DenseNet-121 image classification, patient metadata, Grad-CAM explanations, and Human-in-the-Loop review within a unified diagnostic workflow. The multi-modal model improved accuracy, specificity, F1-score, and ROC-AUC compared with the image-only baseline, although recall decreased. Grad-CAM demonstrated the ability to expose model behaviour across correct and incorrect predictions, while HITL review demonstrated that human intervention could identify and correct model errors.

Overall, the study demonstrates the feasibility of integrating **multi-modal learning, explainability, and human oversight** within a single diagnostic artefact. However, the limited HITL sample, single-dataset evaluation, and absence of external clinical validation restrict the generalisability of the findings. Future work should therefore focus on external validation and larger-scale clinical evaluation.

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